

Nigeria Population Distribution & Gender Demographics Analysis

Project Overview

This project uses QGIS to visualize and analyze population distribution across Nigeria States. Population data stored in a CSV table was integrated with Nigeria's administrative boundary data to produce four choropleth maps showing total population, male population, female population, and the difference between male and female population.

The project demonstrates the integration of tabular data with spatial data, attribute calculation, thematic mapping, and demographic data visualization.

Objective

To examine state-level population distribution across Nigeria and visualize differences in total, male, and female population, including the calculated gender population difference.

Data Used

Nigeria state administrative boundaries - Vector layer

Nigeria population data – Delimited Text layer (CSV Format)

Methodology

1. Imported the Nigeria state administrative boundaries into QGIS.
2. Imported the population data table CSV file as a delimited text layer.
3. Reviewed the population attributes for each Nigerian state.
4. Joined the population table to the corresponding state administrative boundary layer using the appropriate state name field.
5. Created a new attribute representing the difference between male and female population.
6. Calculated the gender population difference directly in QGIS using the field calculator.
7. Applied graduated symbology to create choropleth maps for: Total population, Male population, Female population, Gender population difference.
8. Adjusted classification, labels, and map symbology to clearly communicate the demographic patterns.
9. Created a final print layout containing the four thematic maps.
10. Added essential cartographic elements including titles, legends, north arrows, and scale bars.

Analysis

The four maps provide a visual comparison of population characteristics across Nigerian states and highlight variations in:

- Overall population levels
- Male population distribution
- Female population distribution
- Male–female population differences

Key Output

A four-map choropleth layout presenting the total, male, female, and gender-difference population distributions across Nigeria.

QGIS Techniques

- Importing delimited texts datasets
- Attribute table management
- Table-to-spatial-data integration
- Field calculation
- Choropleth mapping
- Graduated symbology
- Data classification
- Labeling
- Print layout design

Skills Demonstrated

Joining CSV data | Spatial Data Visualization | Demographic Analysis | Tabular Data Integration | Attribute Calculation | Choropleth Mapping | Cartography

Project Significance

This workflow demonstrates how tabular demographic data can be combined with geographic boundaries to produce meaningful spatial visualizations for population analysis, demographic comparison, and location-based decision-making.